Pharmaceutical Reform and Pharmacy Entry: Evidence from Hospitals in China

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Institutional Background

China's pharmaceutical reform aims to eliminate the long-standing "drug mark-up" mechanism in public hospitals.

- **Financing Reform:** Public hospitals previously relied on three revenue sources: service fees, drug mark-ups, and government subsidies. The reform removes drug mark-ups, shifting the system to a two-channel structure: service fees and government subsidies.
- **Price Adjustment:** The policy eliminates drug mark-ups while increasing the prices of medical services, including consultation, surgery, and nursing, to better reflect the value of medical labor.
- **Complementary Policies:** The reform is accompanied by changes in health insurance payment methods and strengthened government responsibility in hospital financing.

Reform Rollout and Scope

The reform was first implemented as a pilot program targeting county-level public hospitals. It introduced a comprehensive set of changes, including reforms to hospital governance, financing mechanisms, personnel and compensation systems, drug procurement and supply, as well as the medical pricing system.

- The pilot was launched in 2012 and covered approximately 300 counties (or county-level cities).
- Following the pilot phase, the reform was gradually expanded nationwide and was fully implemented by the end of September 2017. All public hospitals adopted the reform, with a universal removal of drug mark-ups, except for traditional Chinese medicine herbal decoctions.
- The staggered rollout of the reform provides quasi-experimental variation for empirical analysis.

Distribution of Policy Adoption Years

Policy Year	Frequency	Percent (%)	Cumulative (%)
Never treated (private hospitals)	86,027	34.37	34.37
2012	6,494	2.59	36.97
2013	1,620	0.65	37.62
2014	34,471	13.77	51.39
2015	43,131	17.23	68.62
2016	16,619	6.64	75.26
2017	61,904	24.74	100.00
Total	250,266	100.00	

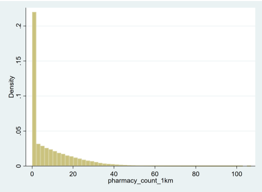
Data Sources

Hospital Data. Information on hospitals is obtained from business registration records, including hospital name, classification (level), industry code, province, county, and geographic coordinates.

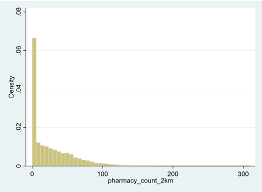
Pharmacy Data. Pharmacy data are collected from Dianping, including merchant name, industry classification, geographic coordinates, and an indicator for chain affiliation.

Reform Data. Information on the timing and implementation of the reform is compiled from official government policy documents and related reports.

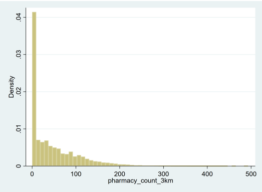
Distribution of Pharmacy Density Around Hospitals



(a) 1km



(b) 2km



(c) 3km

Note: Distribution is highly right-skewed with mass at zero.

Empirical Strategy:TWFE

$$Y_{ict} = \beta_0 + \beta_1 Reform_{ct} + \beta X_{ct} + \delta_i + \delta_t + \epsilon_{ict}$$

- Y_{ict}: Number of pharmacies around hospital i in county c at time t
- $Reform_{ct}$: Policy exposure at the county-year level
- X_{ct}: Control variables at the county-year level
- δ_i: hospital fixed effects
- δ_t: Year fixed effects

Traditional TWFE may be biased under staggered adoption

Empirical Strategy: CS-DID

$$ATT(g, t) = \mathbb{E}[Y_t(1) - Y_t(0) \mid G = g]$$

- g: First year of reform (cohort)
- t: Year
- Compare treated units with not-yet-treated units

- Allows for heterogeneous treatment effects
- Avoids bias from two-way fixed effects under staggered adoption

Estimate dynamic and cohort-specific treatment effects

Summary of TWFE Results

Main Results

- Counts level
  - TWFE estimates: Negative and significant.
  - Including trends: Negative but insignificant.
  - Parallel trend assumption: Not Satisfied
- Log level
  - TWFE estimates: Negative but statistically insignificant.
  - Parallel trend assumption: Not Satisfied
- Indicator(0/1)
  - Estimates: Positive and statistically significant within 2-3 km.
  - Parallel trend assumption: 1 km satisfied,2-3km almost satisfied.

Baseline DID Results:TWFE

OLS

- Outcome: Pharmacy entry around hospitals
- Sample: score=10 | score=8
- Fixed effects: Hospital and year
- Standard errors : clustered at county level

	(1)	(2)	(3)
	pharmacy~1km	pharmacy~2km	pharmacy~3km
did	-0.433**	-0.947*	-1.113
	(0.161)	(0.441)	(0.776)
N	192868	192868	192868
R-sq	0.884	0.895	0.903

Baseline DID Results: TWFE

OLS(log+1)

- Outcome: Log of pharmacy entry around hospitals
- Sample: score = 10 or score = 8
- Fixed effects: Hospital and year
- Standard errors: Clustered at the county level

	(1)	(2)	(3)
	ln(pharmacy $\sim$ 1km)	ln(pharmacy $\sim$ 2km)	ln(pharmacy $\sim$ 3km)
did	-0.028	-0.013	-0.010
	(0.015)	(0.019)	(0.021)
N	192,868	192,868	192,868
R-sq	0.897	0.907	0.911

Extensive Margin: Pharmacy Presence (0/1)

Outcome: Indicator for Any Pharmacy Around Hospitals

	(1) 1km	(2) 2km	(3) 3km
DID	0.007	0.021**	0.018**
	(0.007)	(0.007)	(0.007)
Observations	192,868	192,868	192,868
R^2	0.773	0.728	0.689

Interpretation

- Reform **increases the probability of pharmacy presence** within 2–3 km.

Extensive Margin: Switchers Sample Only

Sample: Hospitals with at least one zero outcome

	(1) 1km	(2) 2km	(3) 3km
DID	-0.020*	0.004	0.000
	(0.008)	(0.008)	(0.009)
Observations	88,111	88,111	88,111
R^2	0.762	0.752	0.725

Interpretation

- Within hospitals that experience entry/exit, the reform **reduces the probability of pharmacy presence within 1 km** by about 2 percentage points.
- No statistically significant effects at 2–3 km.
- Suggests a **local contraction effect** rather than spatial expansion.

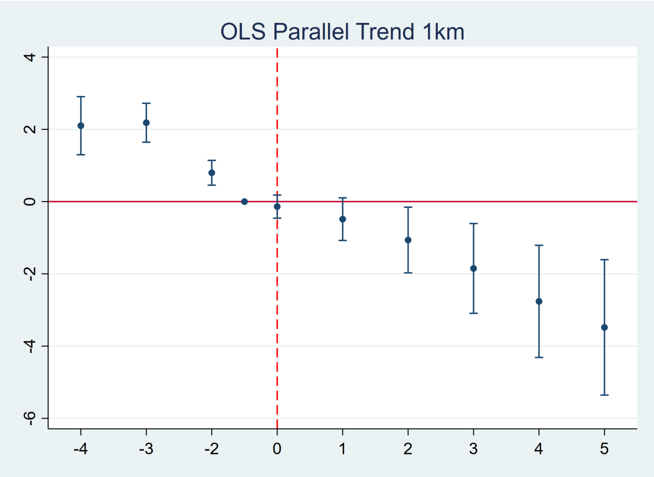
Quantile Results

Quantile Treatment Effects

	Q25	Q50	Q75
1km	-0.419***	-0.072***	-0.087*
	(0.040)	(0.021)	(0.049)
2km	-0.931***	-0.171**	-0.157
	(0.100)	(0.060)	(0.123)
3km	-1.237***	0.206	0.281
	(0.158)	(0.109)	(0.199)
Observations		193,010	

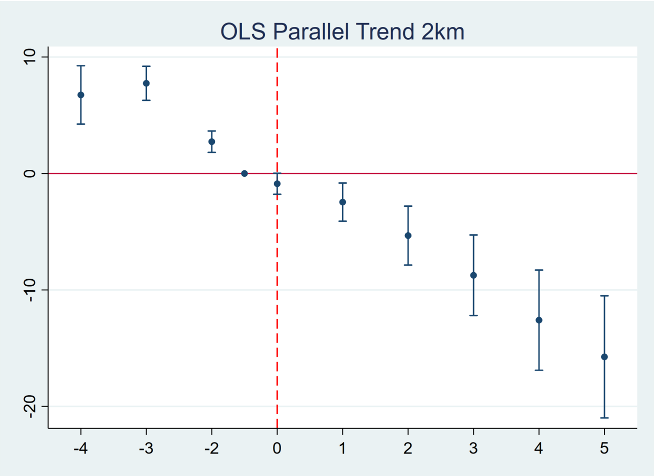
Event Study:TWFE

- Outcome: Pharmacy entry around hospitals (within 1km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



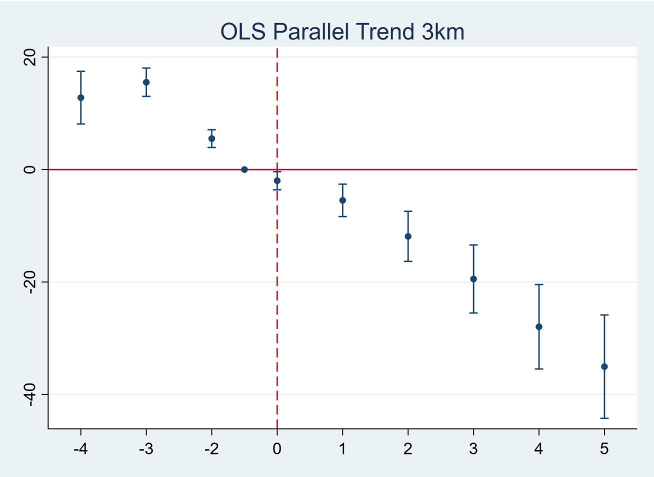
Event Study: TWFE

- Outcome: Pharmacy entry around hospitals (within 2km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



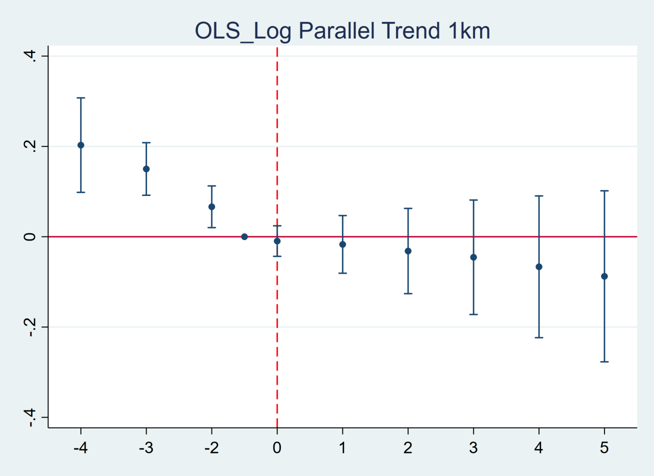
Event Study: TWFE

- Outcome: Pharmacy entry around hospitals (within 3km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



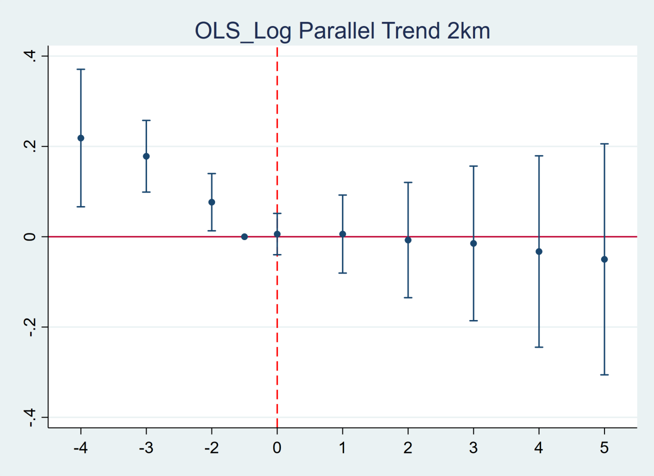
Event Study: TWFE

- Outcome: Log of Pharmacy entry around hospitals (within 1km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



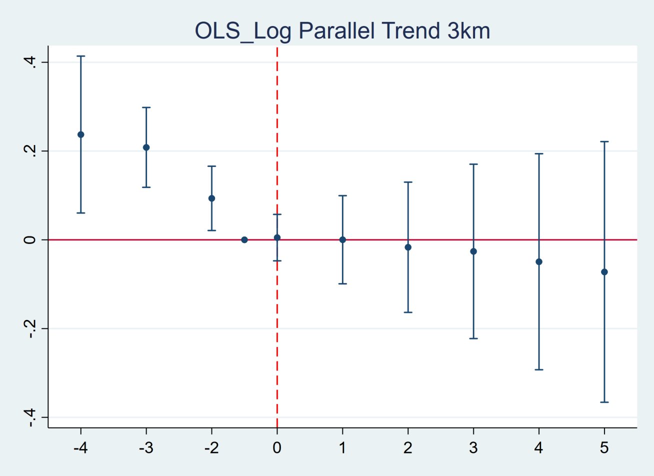
Event Study:TWFE

- Outcome: Log of Pharmacy entry around hospitals (within 2km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



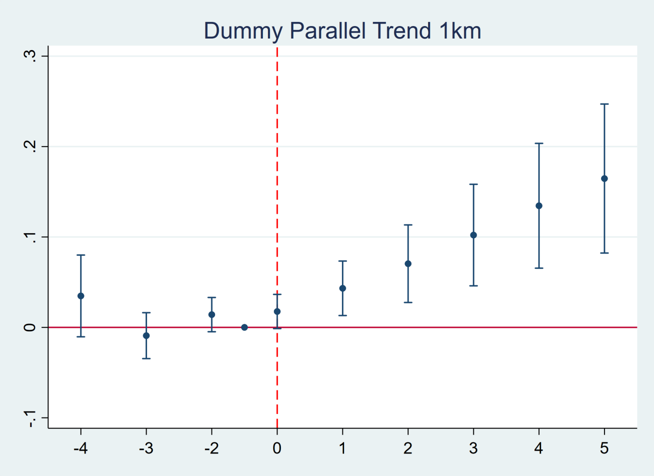
Event Study:TWFE

- Outcome: Log of Pharmacy entry around hospitals (within 3km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



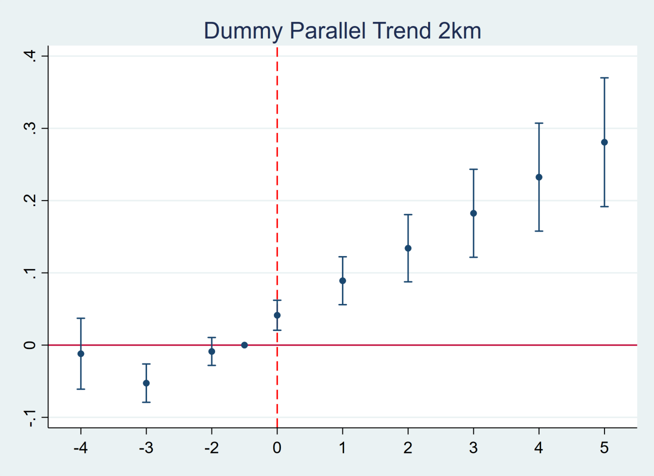
Event Study:Dummy

- Outcome: Indicator of Pharmacy entry around hospitals (within 1km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



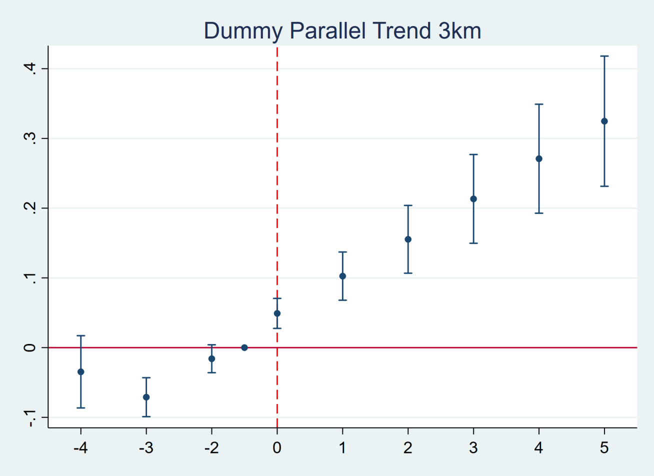
Event Study:Dummy

- Outcome: Indicator of Pharmacy entry around hospitals (within 2km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



Event Study:Dummy

- Outcome: Indicator of Pharmacy entry around hospitals (within 3km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



Summary of CS-DID Results

Key Findings

- Counts level
  - Estimates: Negative and insignificant.
  - Including trends: Negative and insignificant.
  - Parallel trend assumption: Not Satisfied
- Log level
  - Estimates: Positive and significant.
  - Parallel trend assumption: Not Satisfied.
- Dummy
  - Estimates: Positive and significant.
  - Parallel trend assumption: Not Satisfied.

Baseline DID Results: CS-DID

Callaway and Sant'Anna(2021)

- Outcome : insignificant negative
- Control: never treated
- Treatment model: inverse probability
- Standard errors : clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
ATT	-0.074	-0.321	-0.584
	(0.099)	(0.241)	(0.403)
N	124137	124137	124137

Baseline DID Results: CS-DID-Log

Callaway and Sant'Anna(2021)

- Outcome : significant positive
- Control: never treated
- Treatment model: inverse probability
- Standard errors : clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
ATT	0.039 (0.009)	0.059 (0.01)	0.06 (0.01)
N	124137	124137	124137

Baseline DID Results: CS-DID-Dummy

Callaway and Sant'Anna(2021)

- Outcome : significant positive
- Control: never treated
- Treatment model: inverse probability
- Standard errors : clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
ATT	0.029	0.030	0.022
	(0.005)	(0.04)	(0.004)
N	124137	124137	124137

Baseline DID Results: CS-DID

Callaway and Sant'Anna(2021)

- Outcome: insignificant negative
- Control: Not yet Treated
- Treatment model: inverse probability
- Standard errors : clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
ATT	-0.073 (0.097)	-0.348 (0.238)	-0.679 (0.399)
N	124137	124137	124137

Baseline DID Results: CS-DID-Log

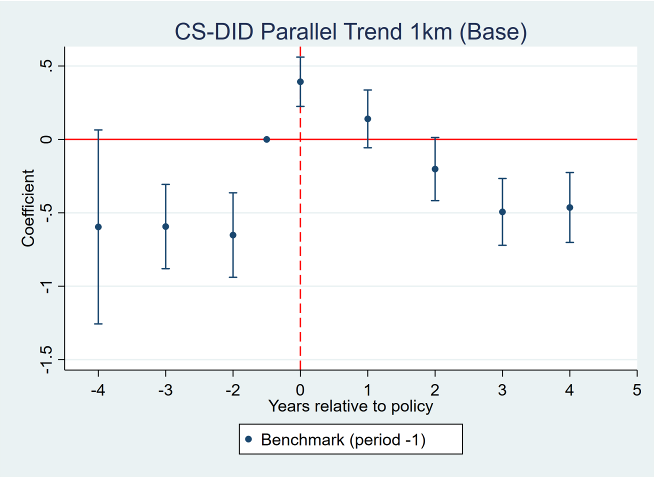
Callaway and Sant'Anna(2021)

- Outcome: significant positive
- Control: Not yet Treated
- Treatment model: inverse probability
- Standard errors : clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
ATT	0.04 (0.009)	0.06 (0.01)	0.63 (0.012)
N	124137	124137	124137

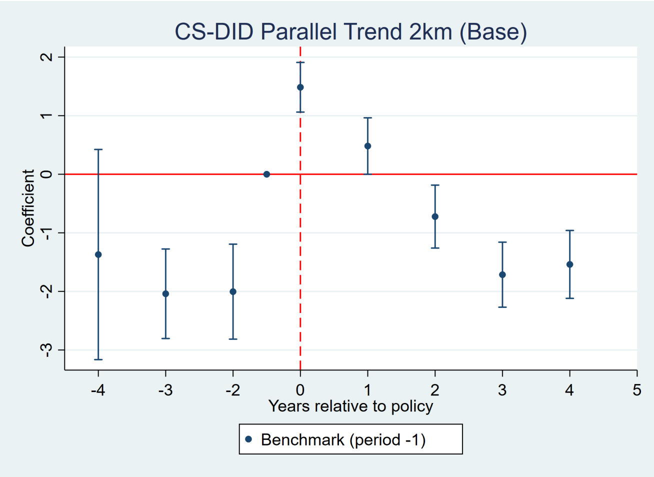
Event Study:CS

- Outcome: Pharmacy entry around hospitals (within 1km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



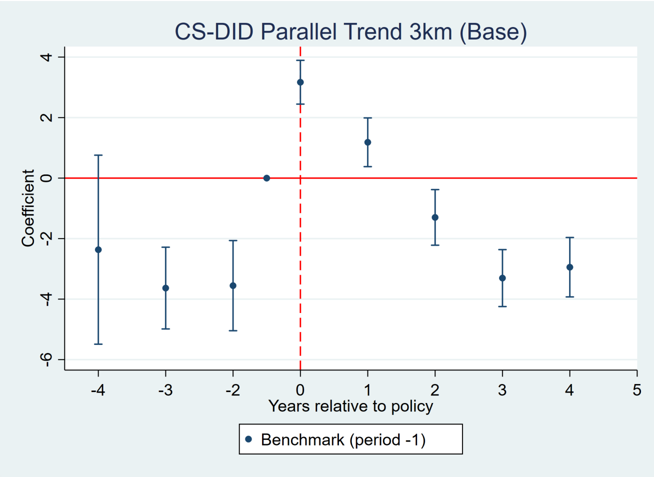
Event Study:CS

- Outcome: Pharmacy entry around hospitals (within 2km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



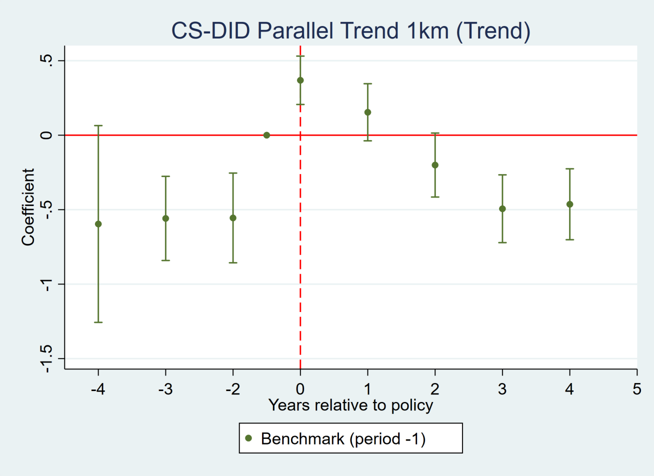
Event Study:CS

- Outcome: Pharmacy entry around hospitals (within 3km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



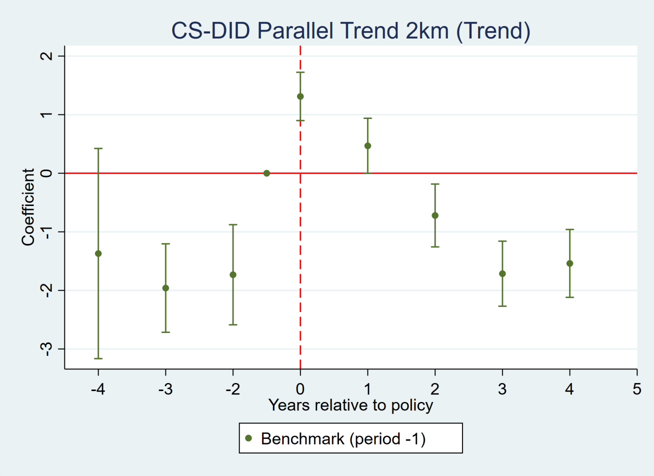
Event Study:CS +trend

- Outcome: Pharmacy entry around hospitals (within 1km)
- Fixed effects: hospital and year
- trend : county year trend
- Standard errors : clustered at county level



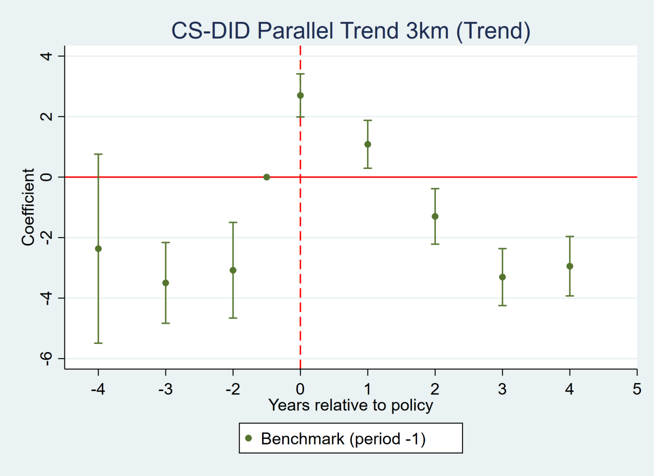
Event Study:CS +trend

- Outcome: Pharmacy entry around hospitals (within 2km)
- Fixed effects: hospital and year
- trend : county year trend
- Standard errors : clustered at county level



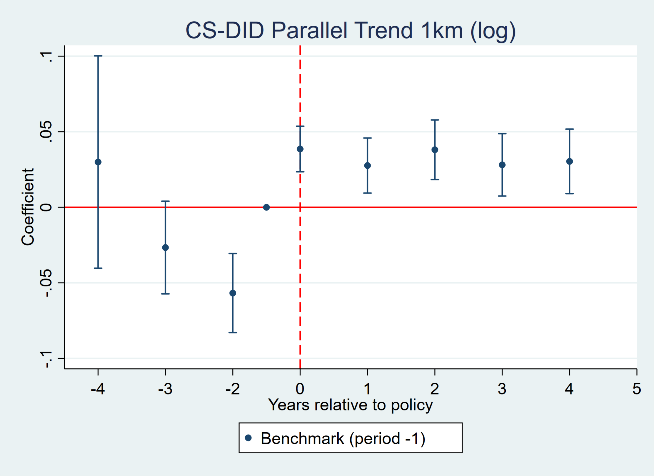
Event Study:CS +trend

- Outcome: Pharmacy entry around hospitals (within 3km)
- Fixed effects: hospital and year
- trend : county year trend
- Standard errors : clustered at county level



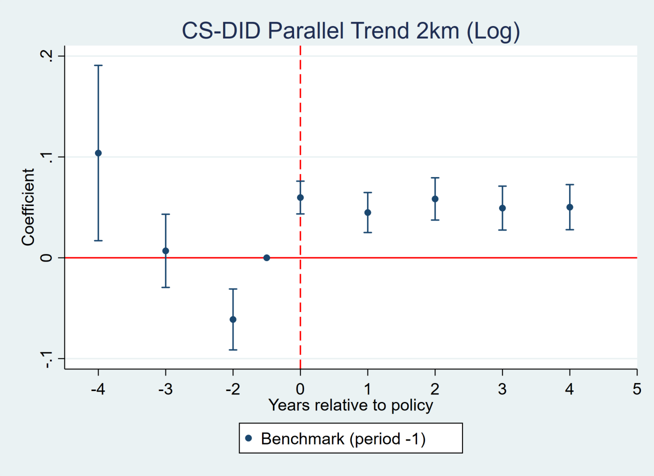
Event Study:CS-Log

- Outcome: Log of pharmacy entry around hospitals (within 1km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



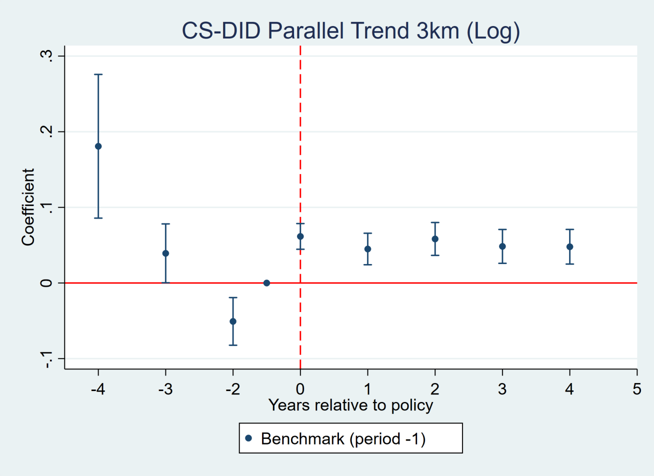
Event Study:CS-Log

- Outcome:Log of pharmacy entry around hospitals (within 2km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



Event Study:CS-Log

- Outcome: Log of pharmacy entry around hospitals (within 3km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



Summary of PPML Results

Key Findings

- PPML estimates: Positive and significant.
- Including trends: Positive and significant.
- Parallel trend assumption: Satisfied

Main Problems

- Coefficient is Positive,while TEFE estimates are negative.

Baseline DID Results: PPML

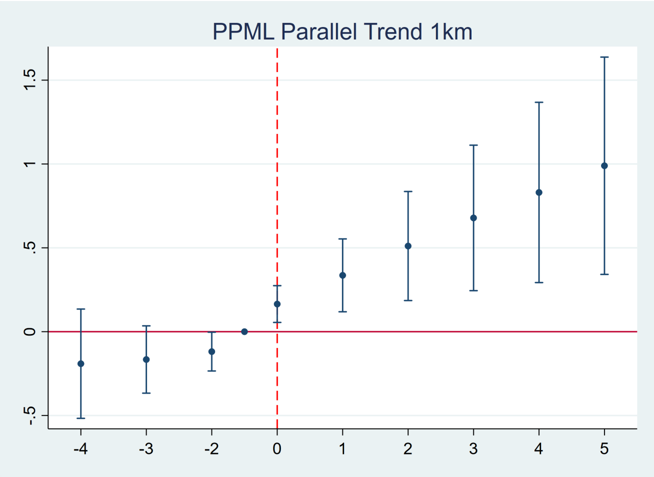
PPML

- Outcome: Pharmacy entry around hospitals
- Fixed effects: hospital and year
- Sample: score=10 | score=8
- Standard errors : clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
did	0.041***	0.047***	0.055***
	(0.016)	(0.015)	(0.016)
N	177459	185543	188251

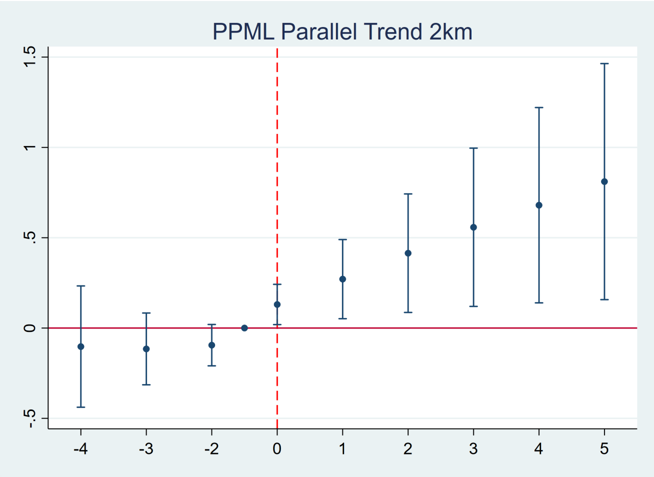
Event Study:PPML

- Outcome: Pharmacy entry around hospitals (within 1km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



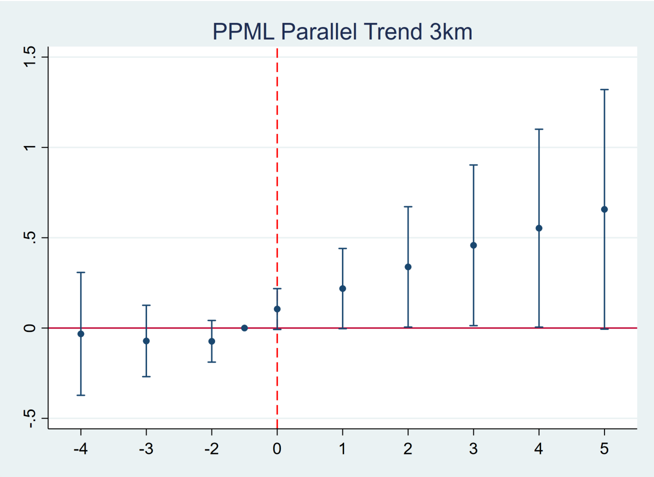
Event Study:PPML

- Outcome: Pharmacy entry around hospitals (within 2km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



Event Study:PPML

- Outcome: Pharmacy entry around hospitals (within 3km)
- Fixed effects: hospital and year
- Standard errors : clustered at county level



Intensive Margin: OLS

OLS Estimates (Positive-Pharmacy Hospitals Only)

Radius	DID Coefficient	Std. Error
1km	+0.457	(0.242)
2km	+1.615*	(0.707)
3km	+2.955*	(1.315)

Interpretation

- Conditional on hospitals already having pharmacy presence, reform is associated with **higher pharmacy counts**.
- Reform may reduce pharmacy entry in marginal/zero-pharmacy areas,
- But increase pharmacy concentration in already-developed markets.

Intensive Margin: CS-DID

Log-OLS Estimates (Positive-Pharmacy Hospitals Only)

Radius	Log Coefficient	Std. Error
1km	+0.609***	(0.015)
2km	+1.286**	(0.398)
3km	+1.239*	(0.709)

Intensive Margin: PPML

PPML Estimates (Positive-Pharmacy Hospitals Only)

Radius	PPML Coefficient	Std. Error
1km	+0.043**	(0.015)
2km	+0.040**	(0.015)
3km	+0.036*	(0.016)

Interpretation

- Among hospitals with sustained positive pharmacy presence, reform increased nearby pharmacy counts on the intensive margin.
- Results suggest reform strengthened pharmacy expansion in existing markets rather than reducing incumbent pharmacy activity.
- Less entry in weak or zero-pharmacy markets,
- Greater concentration where pharmacy ecosystems were already established.

Intensive Margin: Log(1+Y) OLS

Log-OLS Estimates (Positive-Pharmacy Hospitals Only)

Radius	Log Coefficient	Std. Error
1km	+0.038*	(0.016)
2km	+0.049**	(0.017)
3km	+0.044*	(0.018)

Interpretation

- Results closely align with PPML estimates, reinforcing intensive-margin expansion after reform.
- Effects are strongest at 2–3km, suggesting pharmacy growth may have shifted outward beyond immediate hospital frontage.
- Reduced new entry in weak markets。
- Increased pharmacy density in mature local markets.

Intensive Margin: Log(1+Y) CS-DID

CS-DID Estimates (Positive-Pharmacy Hospitals Only)

Radius	**Log Coefficient**	**Std. Error**
1km	+0.070***	(0.010)
2km	+0.088***	(0.0096)
3km	+0.082***	(0.0090)

Mechanism: Chain Pharmacies(OLS)

- Outcome: Chain pharmacies entry around hospitals
- Fixed effects: hospital and year
- Standard errors: clustered at county level

	(1)	(2)	(3)
	Chain pharmacy ∼1km	Chain pharmacy ∼2km	Chain pharmacy ∼3km
did	0.012	0.382	1.017*
	(0.088)	(0.261)	(0.479)
N	188,410	188,410	188,410
R-sq	0.682	0.719	0.733

Mechanism: Non-Chain Pharmacies

- Outcome: Non-chain pharmacies entry around hospitals
- Fixed effects: hospital and year
- Standard errors: clustered at county level

	(1) Non-chain pharmacy ∼1km	(2) ∼2km	(3) ∼3km
did	-0.537***	-1.553***	-2.495***
	(0.117)	(0.310)	(0.538)
N	188,410	188,410	188,410
R-sq	0.828	0.833	0.836

Mechanism: Chain Pharmacies (Share)

- Outcome: Share of chain pharmacies around hospitals
- Fixed effects: hospital and year
- Standard errors: clustered at county level

	(1) Chain share ∼1km	(2) Chain share ∼2km	(3) Chain share ∼3km
did	0.019***	0.021***	0.022***
	(0.004)	(0.004)	(0.004)
N	159,005	170,604	174,613
R-sq	0.788	0.840	0.851

Mechanism: Chain Pharmacies (PPML)

- Outcome: Chain pharmacies around hospitals (count)
- Estimator: PPML with hospital and year fixed effects
- Standard errors: clustered at county level

	(1) Chain pharmacy ∼1km	(2) Chain pharmacy ∼2km	(3) Chain pharmacy ∼3km
did	0.334***	0.353***	0.358***
	(0.088)	(0.082)	(0.079)
N	139,926	149,752	153,318

Mechanism: Non-chain Pharmacies (PPML)

- Outcome: Non-chain pharmacies around hospitals (count)
- Estimator: PPML with hospital and year fixed effects
- Standard errors: clustered at county level

	(1) Non-chain ∼1km	(2) Non-chain ∼2km	(3) Non-chain ∼3km
did	0.007	0.010	0.016
	(0.013)	(0.013)	(0.013)
N	171,807	180,668	183,617

Mechanism: Chain Pharmacies (Share, PPML)

- Outcome: Share of chain pharmacies around hospitals
- Estimator: PPML with hospital and year fixed effects
- Standard errors: clustered at county level

	(1) Chain share ∼1km	(2) Chain share ∼2km	(3) Chain share ∼3km
did	0.205***	0.229***	0.242***
	(0.061)	(0.057)	(0.057)
N	129,869	141,824	146,097

Summary

- DID: OLS negative and significant ,CS negative but insignificant ,PPML positive and significant
- Event Study: PPML, supporting causal inference
- How to adjust event study

The reform increases pharmacy entry around hospitals